The Central Limit Theorem: Binomial Distribution

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In this study, we aim to explore the Central Limit Theorem by analyzing the normality of sample means generated from a binomial distribution and understand the necisary sample size to have a high level of confidence in our statistical infernece. We will utilize the Shapiro-Wilk Normality Test to assess the means of samples of different sizes and compare that to the probability of a success when generating our means, and find how these different factors affect normality. In this testing, we found that both the sample size of means and the probability we gave our generation function have an effect on the normality of our samples. We found that as we increased the sample size, our samples more closely resembled a normal distribution, reinforcing the importance of having sufficient repetitions for reliable statistical influence. We also found that as the probability approached .5, we needed a smaller sample size to have a similar confidence in the sample being normal. This means that when dealing with catagorical data, as our probability gets further from being even, we need a larger sample size to have similar confidence in our inference.

Table of contents

Introduction **3**
- Background . . . 3
  - Shapiro-Wilkes Test . . . 3
  - Binomial Distribution . . . 3
- Expectations . . . 4
- Data . . . 4

Methodology **5**
- Exploratory Data Analysis . . . 5
- Analysis . . . 6

Conclusions **7**
Future Recommendations 7

References **8**

Appendix **9**
Appendix 1 9
Appendix 2 9
Appendix 3 10

Introduction

The Central Limit Theorem is an important concept to understand when trying to recognize the necessary amount of data needed to make a statistical inference. This Theorem is essential in statistical inference because it provides the basis for hypothesis testing, confidence intervals, and other methods used for analyzing data.

In this project, we are investigating the behavior of sample means drawn from the binomial distribution and and assess their normality using the Shapiro-Wilkes Normality Test. By simulating repeated sampling processes, we aim to evaluate how sample size (n) and the probability of the items in that sample being "TRUE" (p) influence the test's outcomes.

Our approach involves creating data sets with different sample sizes and probabilities, calculating their means, and assessing their normality. This process aims to evaluate the reliability of the Central Limit Theorem under varying conditions. We anticipate that larger sample sizes will more closely approximate the normal distribution, consistent with the theorem's expectations. We also believe that increasing the probability, up to 0.5, will reduce the necessary sample size needed to maintain our confidence in the normality of our data.

In this report, we will present how we went about generating our data. We will then perform the exploratory data analysis, where we will obtain insights into the data and understand the effects of our factors. These effects will help us to understand how the Central Limit Theorem reacts to different conditions and demonstrate the importance of sample size in drawing statistical conclusions.

Background

Shapiro-Wilkes Test

The Shapiro-Wilkes test is a method used to test the normality of a data set. The test statistic is called W and is a number between 0 and 1. W values closer to 1 indicate a data set that appears to be "more normal." We will run this test on each combination of n and p for every experiment. The p-value shows the probability of observing W given the null hypothesis.(Navarro 2022)

Null Hypothesis: A set of numbers is normal. Alternative Hypothesis: A set of numbers is not normal.

Binomial Distribution

The binomial distribution is a simple distribution of the result of multiple experiments that are independent and have two outcomes. The probability of a success or "TRUE" for each number is given by the probability "p". The most common example for a probability of 0.5

is the results of flipping multiple coins and measuring whether or not it landed on heads. An example for a binomial test for a probability of 0.1 is to draw a single marble out of a bag where the bag contains 9 red and 1 blue marble and record whether or not you draw the blue marble.

One rule of thumb the Binomial Distribution is that when we group the data by successes and failure we should have a minimum of 10 in each group for the Central Limit Theorem to hold. Mathematically, $n * p > 10$ and $n * (1 - p) > 10$.(Anonymous, n.d.)

Expectations

The Central Limit Theorem suggests that as the number of samples (N) increases, the distribution of the sample means will approach a normal distribution, regardless of the distribution of the original data. Based on this, we expect that:

1. Larger samples (n) will produce more sample distributions that closely follow a normal distribution, leading to higher Shapiro-Wilkes W statistics and larger p-values.
2. As our probability (p) approaches 0 or 1 we will need a greater sample size to get similar W statistics and p-values.

Null Hypothesis: When $n * p > 10$ and $n * (1 - p) > 10$ then the Central Limit Theorem Holds for the Binomial Distribution.

Alternative Hypothesis: The Central Limit Theorem does not hold when $n * p > 10$ and $n * (1 - p) > 10$.

Data

To generate our data, we start with a list of numbers indicating the desired sample sizes n. We then have a number indicating the number of sample means we want in each group N. We also give a list of probabilities indicating how much we favor one outcome p. We loop through each possible combination of the items in p and n. Finally, we have a single number that represents the number of replicates of each combination N_Exp. For this experiment:

- **Sample Sizes (n)**: A vector of all numbers from 5 to 500.
- **Number of Samples (N)**: One repetition count of 30.
- **Probabilities of "Success" (p)**: A vector of 5 numbers from 0.1 to 0.5 by 0.1.
- **Number of Experiments (N_{Exp})**: One thousand iterations of all combinations

We first multiply the n by N and N_exp, and we create a list of that length using the `rbinom` function where each item has a p probability of being "TRUE". We turn that list into a matrix with n rows and find the mean of each row. We then take those means and turn them into a

matrix with N rows and run a Shapiro-Wilkes test on each row and record the n,p,p_vl, and W for each and put it into a database with all of the other combinations.

Methodology

To analyze the data we generated, we first interpret all the p-values from the Shapiro-Wilkes normality tests using $\alpha = 0.1$. If the p-value is greater that 0.9, we assume the sample distribution is normal. We compare the proportion of normal sample distributions for each n by calculating the proportion of p-values over 0.9.

Exploratory Data Analysis

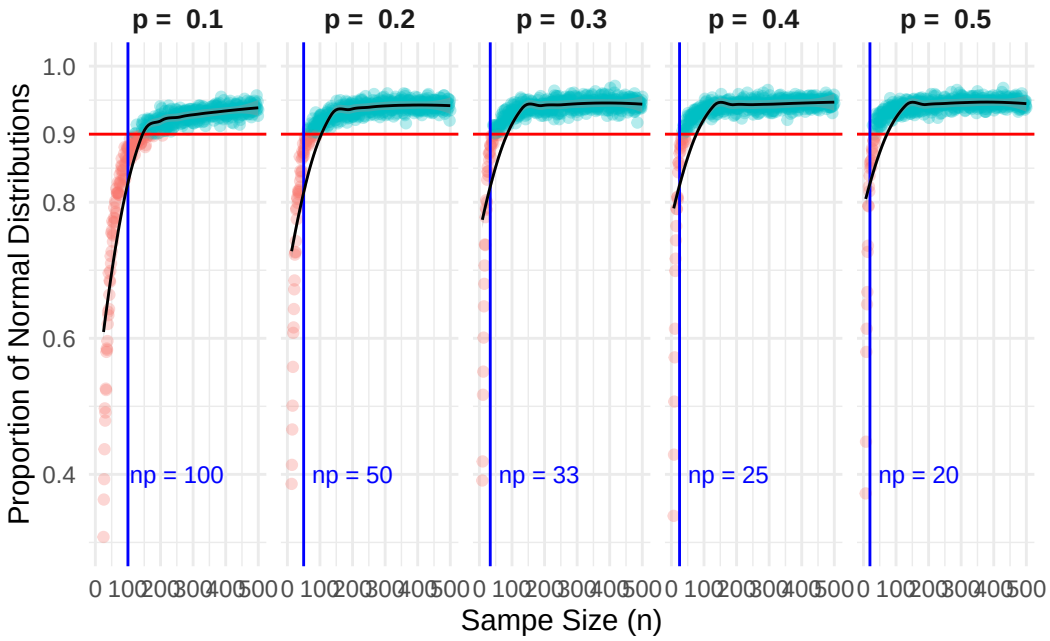


Figure 1: Fig 1: Scatter plot indicating as p and n increase, the proportion of normal sample distributions also increase.

In Fig 1 we plot the proportion of normal sample distributions against the sample size for each p. We also plot a vertical line representing our rule of thumb value for each p and a horizontal line indicating where our proportions are over 0.9. We see that as the sample size of n increases the proportion of normal distributions also increases for every p. Additionally as p increases, the proportions associated with smaller sample sizes increase and we have far fewer proportions below our 0.9 threshold.

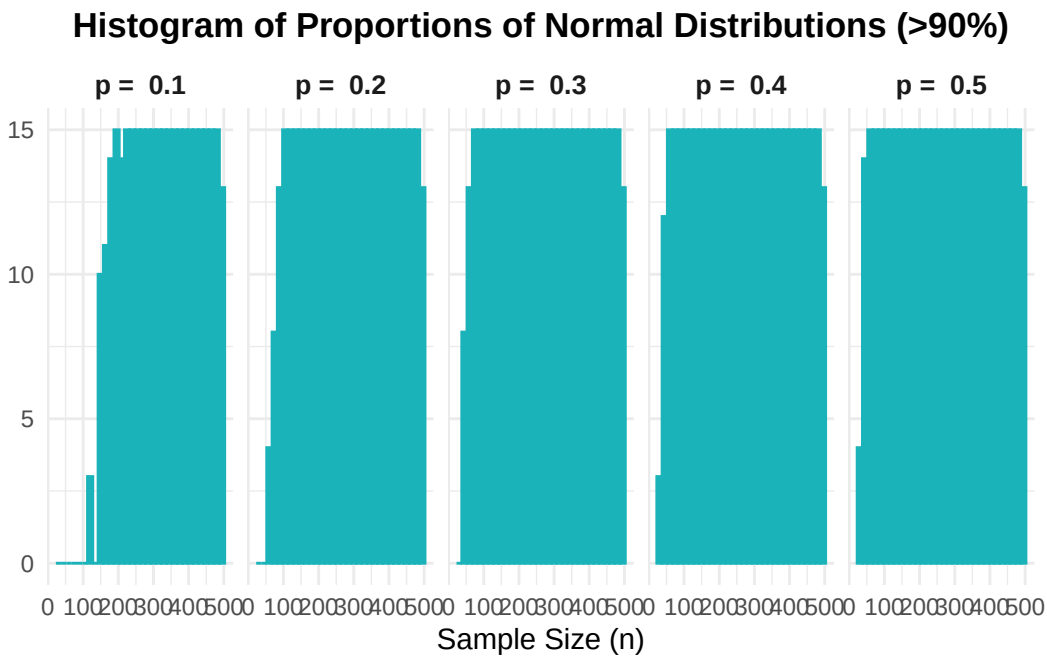


Figure 2: Fig 2: Histogram showing that as p increases, the distribution of normal distributions become more uniform.

Fig 2 is a Histogram showing the distributions of proportions of normal distributions above 90% by n for each p. Similar to Fig 1, as p and n both increase, the proportion of normal distributions increases. You can see that as p increases, the proportions are more consistently over 90%.

Analysis

As previously mentions the rule of thumb is that our sample size times our proportion should be greater than ten in order for the Central Limit Theorem to hold. This translates to:

-n=150 for a p=0.1 -n=50 for a p=0.2 -n=33 for a p=0.3 -n=25 for a p=0.4 -n=20 for a p=0.5

Our data challenges this rule of thumb. We observe that when our p is low we need significantly larger sample sizes than we assumed to be confident that our data is normal. when p is larger, we see a larger proportion of normal data but at our assumed sample size, it is still much lower than we would hope for.

Previously we made the assumptions that 1) Larger samples (n) will produce more sample distributions that closely follow a normal distribution. And 2) As our probability (p) approaches 0 or 1 we will need a greater sample size to get similar W statistics and p-values. These

assumptions held true throughout our testing. Our findings were inconsistent with our Null Hypothesis that When $n * p > 10$ and $n * (1 - p) > 10$ then the Central Limit Theorem Holds for the Binomial Distribution. We found that we needed larger sample sizes than this to be confident that our data was normal. For this reason, we reject our Null Hypothesis in favor of our alternative that, The Central Limit Theorem does not hold when $n * p > 10$ and $n * (1 - p) > 10$.

Conclusions

When sampling from a Binomial distribution, we should use larger sample sizes than $n*p > 10$ and $n * (1 - p) > 10$. From observations in our data, we would recommend sample sizes of approximately:

-n=200 for a p=0.1 -n=150 for a p=0.2 -n=120 for a p=0.3 -n=100 for a p=0.4 -n=70 for a p=0.5

This ensures that the Central Limit Theorem will hold a vast majority of the time and will help us to be confident that any insights we gain from the data are not misrepresented by a small sample size.

Future Recommendations

To further explore the robustness of the Central Limit Theorem, future research might involve generating data from different underlying distributions. Additionally, it would be beneficial to use other normality tests alongside the Shapiro-Wilkes test to assess normality across methods. Examples of these tests include the Kolmogorov-Smirnov test and the Anderson-Darling test. Our methodology can also be improved by performing more technical statistical analyses to pinpoint exact recommendations for sample sizes of the Binomial Distribution.

References

Anonymous. n.d. "The Binomial Distribution." https://math.libretexts.org/Courses/University_of_St._Thomas/Math_101%3A_Finite_Mathematics/03%3A_Probability_Distributions_and_Statistics/3.03%3A_The_Binomial_Distribution.

Navarro, Danielle. 2022. "Checking the Normality of a Sample." https://stats.libretexts.org/Workbench/Learning_Statistics_with_SPSS_-_A_Tutorial_for_Psychology_Students_and_Other_Beginners/10%3A_Comparing_Two_Means/10.08%3A_Checking_the_Normality_of_a_Sample.

Appendix

Appendix 1

Tables of Raw Data. First and last five observations are shown.

n	N	p	p_vl	W	normal
5	30	0.1	2.00e-07	0.6317536	FALSE
5	30	0.1	0.00e+00	0.5148997	FALSE
5	30	0.1	0.00e+00	0.5181304	FALSE
5	30	0.1	2.00e-07	0.6279799	FALSE
5	30	0.1	7.84e-05	0.8042396	FALSE
5	30	0.1	9.00e-07	0.6827558	FALSE

	n	N	p	p_vl	W	normal
2479995	500	30	0.5	0.0860620	0.9391073	TRUE
2479996	500	30	0.5	0.5545975	0.9705595	TRUE
2479997	500	30	0.5	0.2341642	0.9553216	TRUE
2479998	500	30	0.5	0.9990773	0.9930222	TRUE
2479999	500	30	0.5	0.0517284	0.9308511	TRUE
2480000	500	30	0.5	0.0500405	0.9303092	TRUE

Appendix 2

Functions for Generating Data.

```
main <- function(n, N, N_Exp, s=1, p=.5) {n  results<-data.frame(n=integer(),n
```

```
proj1<-function(n, N, N_Exp,p){
  if(!exists("main", mode = "function")){
    source("Binom_main.R")
  }
  results<-data.frame(n=integer(),
                      N=integer(),
                      p=numeric(),
                      p_vl=numeric(),
                      W=numeric()
  )
  for (i in N){
```

```
    for (j in n){
      for (k in p)
      results<-rbind(results,main(j,i,N_Exp,p=k))
    }
  }
  results<-data.frame(results)
  return(results)
}
```

Appendix 3

R Code.

```
knitr::opts_chunk$set(echo = TRUE)

library(tidyverse)
library(knitr)
knitr::write_bib(c('knitr', 'stringr', 'dplyr', 'tidyverse'), file = 'pack.bib')
if(!exists("proj1", mode = "function")){
  source("Binom_proj1.R")
}
n<-c(5:500)
N<-c(30)
N_Exp<-c(1000)
p<-c(0.1,0.2,0.3,0.4,0.5)
start_time <- proc.time()
testmc <- proj1(n,N,N_Exp,p=p)
end_time <- proc.time()

timetorun<-start_time - end_time

testmc<-testmc|>mutate(normal=p_vl>=0.05)

testbj<-testmc|>
  group_by(n,p)|>
  summarise(
    Prop=mean(normal))|>
  mutate(normal=Prop>=.9)

testbj2<-testbj|>
  filter(Prop>=.9)
```

```r
ntimesp<-data.frame(p=unique(testbj$p),np=10/unique(testbj$p))

ggplot(testbj,aes(x=n,y=Prop))+
  geom_point(aes(color=normal),alpha=0.3,show.legend = F)+
  geom_hline(yintercept=.9,color="red",linewidth=.5)+
  geom_vline(data=ntimesp,aes(xintercept=np),linewidth=.5,color="blue")+
  geom_smooth(method="loess",linewidth=.5,color="black")+
  geom_text(data=ntimesp,aes(x=np+150,y=.4,label=paste("np =", round(np,0))),size=3,color="bl

  facet_wrap(~p, nrow=1,labeller=as_labeller(function(x) paste("p = ",p)))+
  labs(y="Proportion of Normal Distributions",
       x="Sampe Size (n)")+
  scale_y_continuous(limits=c(.3,1),breaks=c(.4,.6,.8,.9,1))+
  theme_minimal()+
  theme(strip.text=element_text(face="bold",size=10))
ggplot(testbj2,aes(x=n)) +
  geom_histogram(binwidth=15,fill="#1AB3B9",color="#1AB3B9") +
  facet_wrap(~p,nrow=1,labeller=as_labeller(function(x) paste("p = ",p)))+
  theme_minimal()+
  labs(title="Histogram of Proportions of Normal Distributions (>90%)",x="Sample Size (n)")+
  theme(strip.text=element_text(face="bold",size=10),
        plot.title=element_text(hjust=.5, face="bold"),
        axis.title.y=element_blank())

kable(head(testmc))
kable(tail(testmc))
cat(readLines("Binom_main.R"), sep = "n")
cat(readLines("Binom_proj1.R"), sep = "\n")
```